Tyler Yan

(416) 529-6733 | tylery202@gmail.com | linkedin.com/in/tyler–yan | github.com/tyyan03 | tyyan03.github.io

Education

University of Toronto — **Expected Graduation: April 2026**
Bachelor of Applied Science in Computer Engineering | Minor in Artificial Intelligence — *Toronto, ON*

- Dean's List: 1st, 2nd, 3rd, and 4th year | GPA: 3.69/4.00
- Courses: Software Design and Communication, Applied Fundamentals of Deep Learning, Operating Systems, Algorithms and Data Structures, Intro. to Machine Learning, Intro. to Artificial Intelligence, Computer Architecture

Work Experience

MannLab — **May 2025 – April 2026**
Systems Engineer (XR Assistive Technology) | GitHub Page — *Toronto, ON*

- Led a 4-person team under Prof. Steve Mann to develop an augmented reality navigation aid for visually impaired users with C# and Unity, deployed on the Meta Quest headset; awarded Certificate of Recognition as a top research project
- Achieved 91.7% collision avoidance in user trials by developing on-device machine learning models for obstacle detection and safe path segmentation, delivering multimodal (haptic, audio, and visual) feedback within 200ms
- Coordinating with accessibility service organizations and experts to align system features with real-world user needs

ePIC Blockchain Technologies — **May 2024 – Aug 2025**
Software Engineer Intern — *Toronto, ON*

- Achieved a 100% reduction in firmware validation test setup errors and inconsistent results by developing Autotest, a Rust/Python/YAML reliable testing framework to automate quality assurance and standardize pre-release verification
- Reduced MCU board debug time by 60% by creating a hardware validation application in Rust with automated boot-time tests and detailed failure diagnostics, improving clarity and reducing false positives for manufacturing technicians
- Improved system startup stability and mining efficiency by 35% through researching optimal temperature thresholds to prevent initialization in suboptimal conditions that caused dead chips, reduced hashrates, and startup failures
- Built a full-stack miner performance tracker using TypeScript, HTML/CSS, and Python with a FastAPI backend and SQL database, enabling users to monitor, visualize, export, and compare real-time data via a user-friendly interface
- Redesigned the miner dashboard UI using TypeScript, HTML, and CSS based on customer feedback, significantly improving miner management efficiency and monitoring transparency, resulting in high client satisfaction
- Mentored an intern for 4 months, providing structured onboarding, pair programming sessions, and code review support

Personal Projects

DePhi 0DTE — **Apr 2025 – Jan 2026**

- Led 4 software engineers to build a decentralized options trading platform on the Aptos blockchain, enabling 100% on-chain order placement, execution, and settlement, and reducing trading fees to almost zero via smart contracts
- Designed an intuitive frontend using TypeScript/HTML/CSS with real-time blockchain price feeds, wallet authentication, and position management tools, improving accessibility and usability for non-technical users
- Enabled low-latency (<1s), high-frequency trading while ensuring transparency and security by deploying gas-optimized Move smart contracts

SafeStride | *Source Code* | *Project Page* — **Dec 2024 – Jan 2025**

- Improved mobility safety and accessibility for visually impaired users by designing a two-device wearable system that translated visual input into directional haptic feedback and audio alerts for obstacles
- Developed the real-time obstacle detection system using OpenCV and YOLOv8, achieving a <100ms image-to-alert response time by optimizing Flask server-client communication across two mobile devices

Leadership

University of Toronto - New College Don (Residence Assistant) — **Aug 2023 – Present**

- Responsible for a community of 100+ students, ensuring their safety, well-being, and social development
- Increased engagement in academic and social initiatives by 25% year-over-year by providing one-on-one mentorship and creating monthly programming strategies focused on mental health awareness and social connection

Technical Skills & Interests

Languages: Rust, Python, C, C++, C#, Java, SQL, HTML5, CSS, TypeScript, JavaScript, Assembly, Swift
Developer Tools: Git, Linux, Unix, Docker, ReactJS, PyTorch, Pandas, MongoDB, NumPy, Flask, Bootstrap, scikit-learn
Interests: Basketball, Songwriting, Guitar, Chess, Juggling, Classic Rock, Sketching